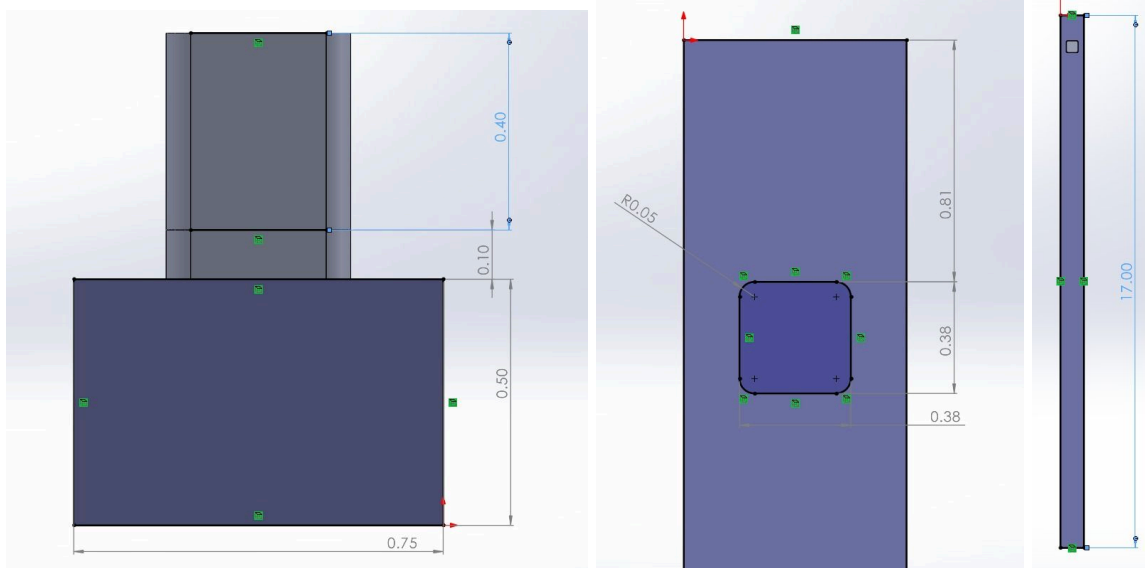


5.2 Your Design, Upload to Portfolio

5.2.1 Results

1. Images of CAD model. Must show all key dimensions



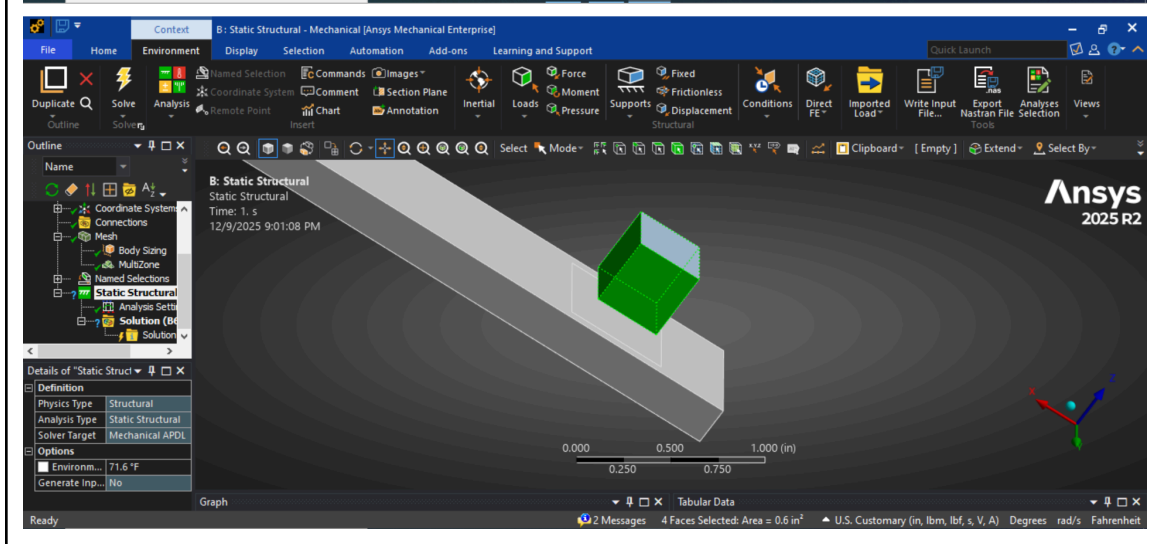
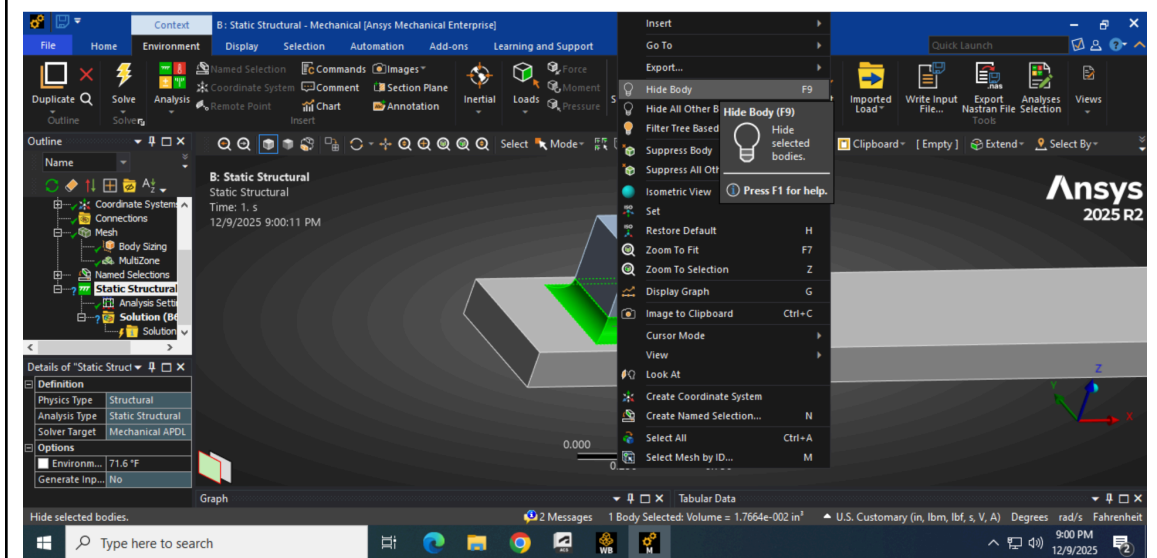
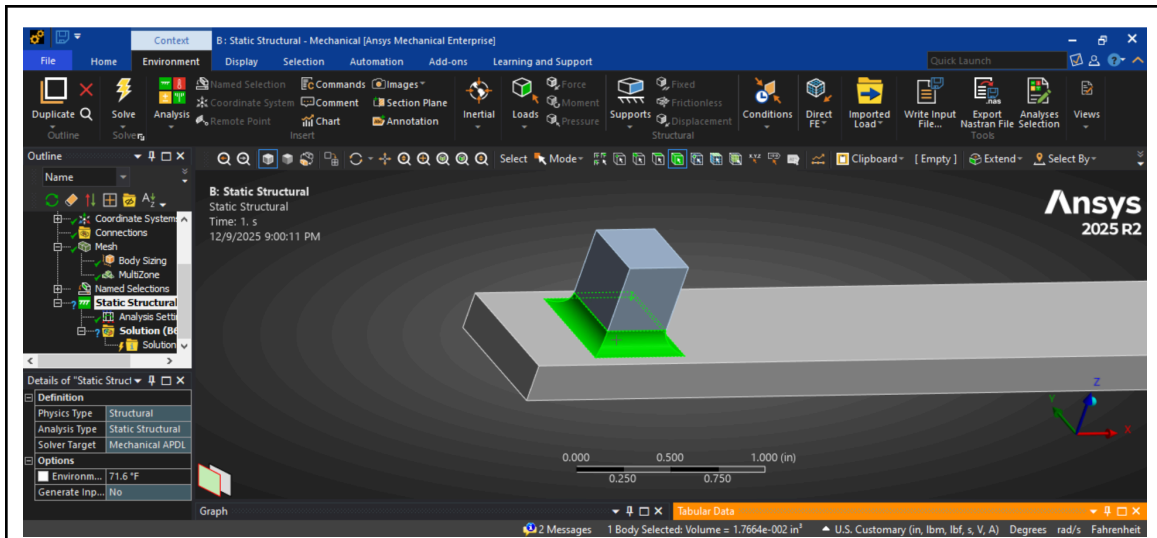
2. Describe material used and its relevant mechanical properties

Our final design uses High Alloy Steel, AerMet 100 (solution treated and aged). This material was chosen because it is both strong and durable and allows the handle dimensions to be reduced for higher strain sensitivity while obtaining all required safety factors. Its mechanical properties are: Young's modulus = 28×10^6 psi, Poisson's ratio = 0.30, yield/tensile strength = 235 ksi, fracture toughness = 91,000 $\text{psi}\sqrt{\text{in}}$, and fatigue strength at 10^6 cycles = 135 ksi. These properties ensure that the design meets and exceeds the requirements for yield, fracture, and fatigue safety margins.

3. Diagram communicating how loads and boundary conditions were applied to your FEM model.

In Ansys mechanical, the boundary conditions and loads can be applied in the static structural tab. Figure 1 shows the streamline process of adding the clamped boundary condition by hiding the fillet body, selecting the faces of the drive, and adding a displacement boundary of 0 at each face for each direction. Figure 2 shows the process of adding the load given 600 lb-in torque at the end of the handle. Note that the applied load is $600/8 = 75$ lbf.

Figure 1: Adding Clamped Boundary



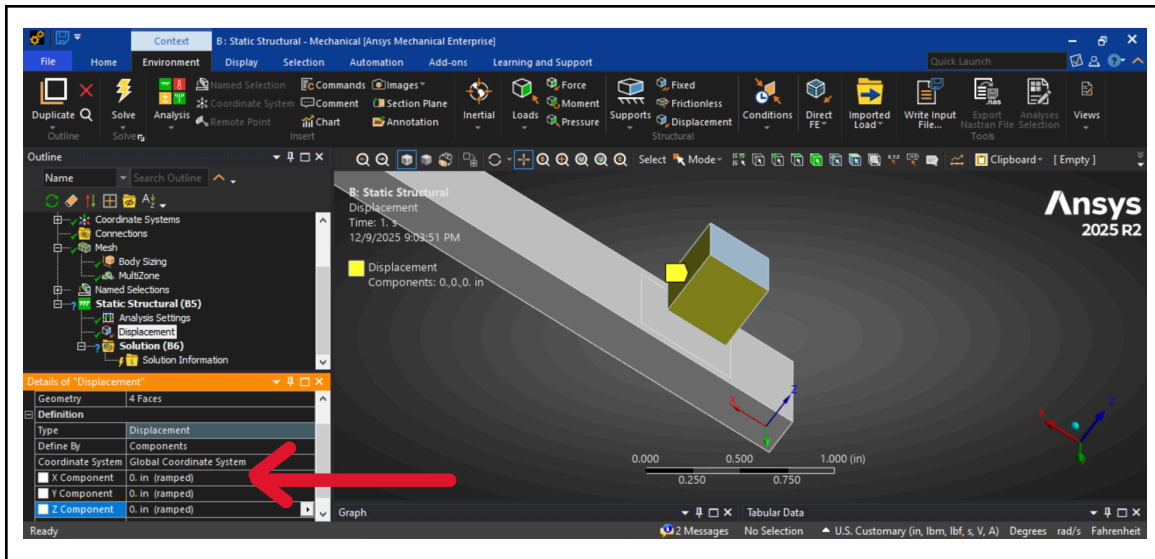
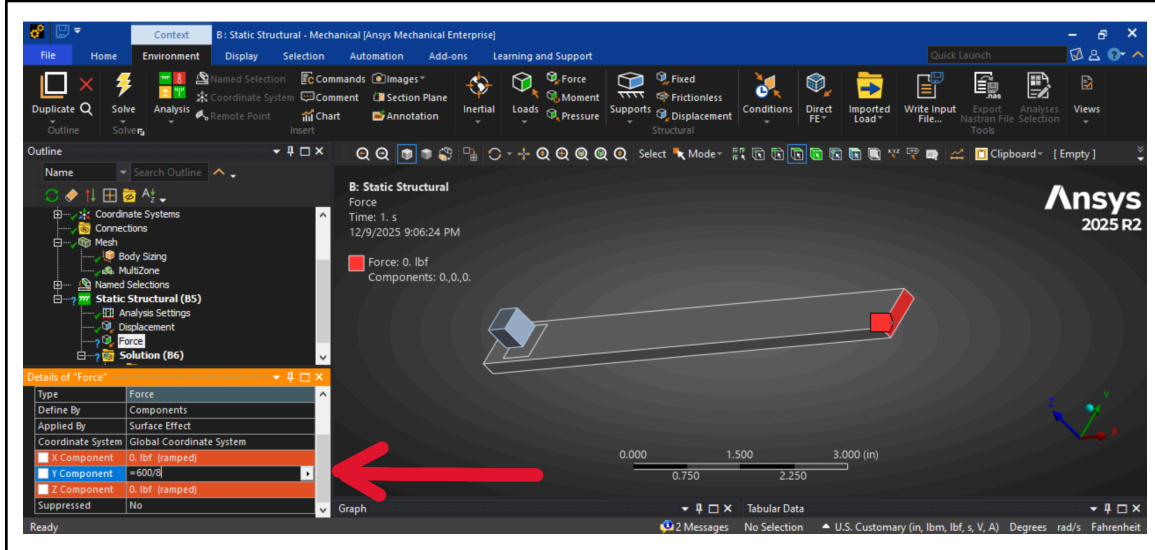


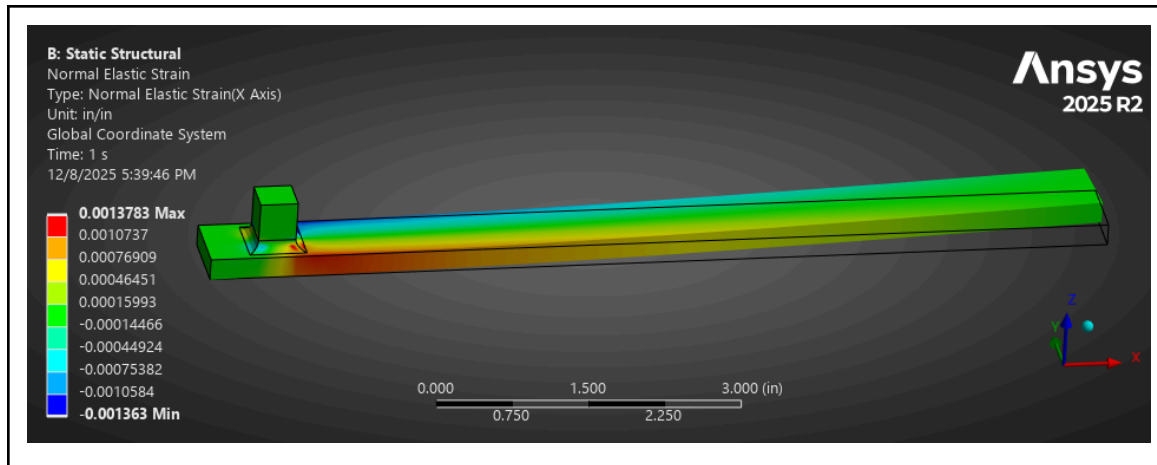
Figure 2: Adding Load to End of Handle



4. Normal Strain contours (in the strain gauge direction) for FEM

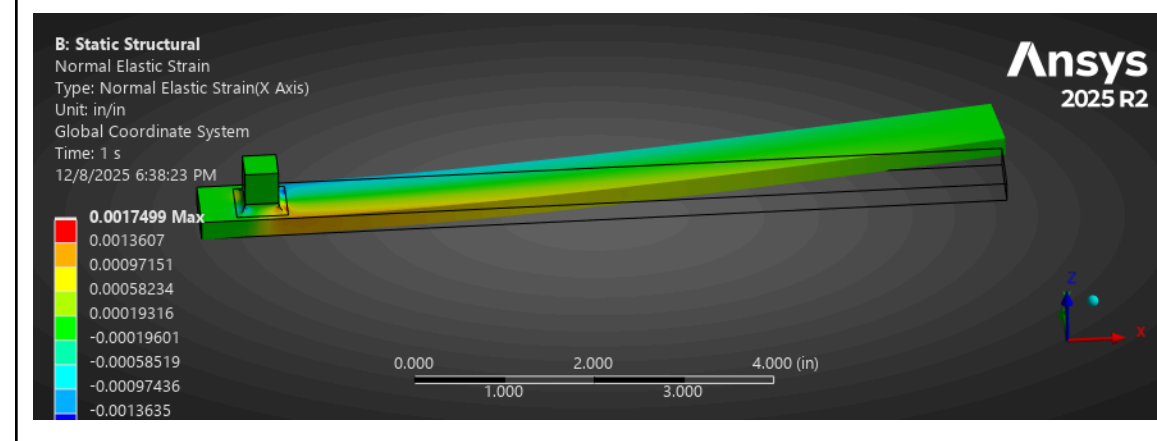
The contour plot for normal strain shown in Figure 3 was obtained after using a mesh size of 0.25 in for the handle and 0.06 in for both of the drive components. Note that the direction of the strain gauge is along the x-axis (not the z-axis originally provided by the baseline design)

Figure 3: Normal Elastic Strain (X Axis) PRE Mesh Refinement



After refining the original mesh size to 0.1in and 0.02 in for the handle and drive components respectively, the contour plot for normal elastic strain was obtained and is shown in Figure 4.

Figure 4: Normal Elastic Strain (X Axis) POST Mesh Refinement

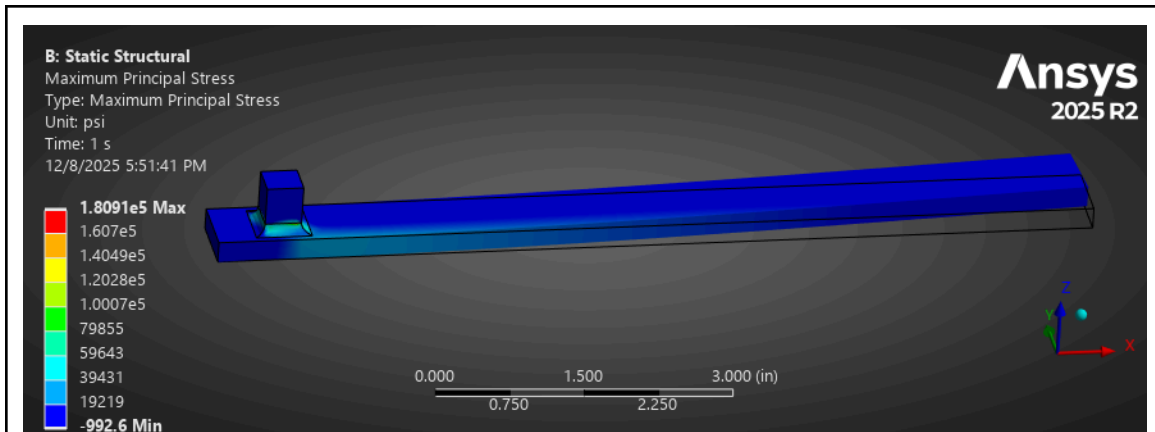


5. Contour plot of maximum principal stress for FEM

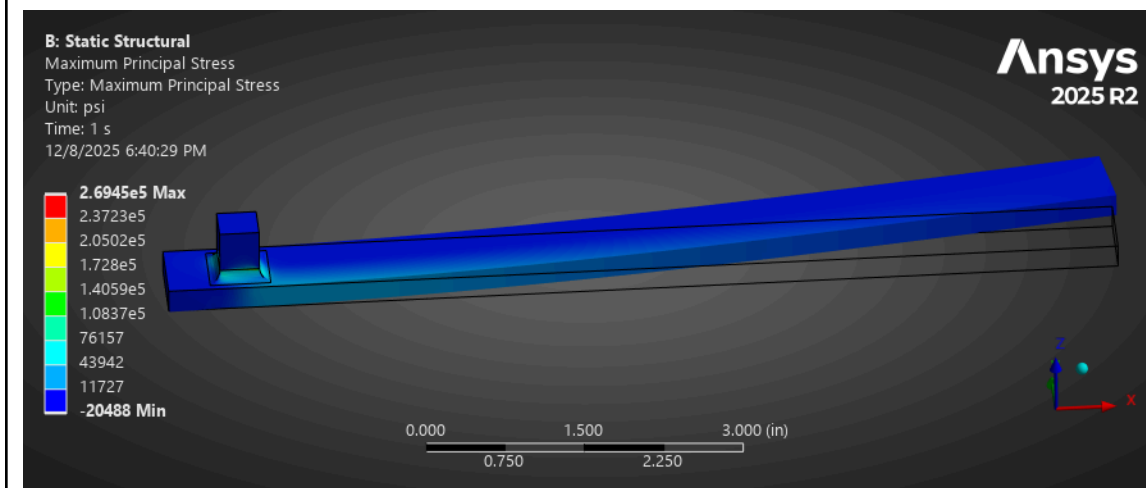
The following are contour plots of the max principal stress, which is seen to be concentrated around the fillet of the drive as well as near the strain gauge location

Figure 5: Maximum Principal Stress

Pre Mesh Refinement



Post Mesh Refinement



- Summarize results from FEM calculation showing maximum normal stress (anywhere), load point deflection, strains at the strain gauge locations

Figure 6: Maximum Strain Values Over Time

Pre Mesh Refinement

Post Mesh Refinement

Maximum Value Over Time	
Normal - X Axis	1.057e-003 in/in
Normal - Y Axis	-3.172e-004 in/in
Normal - Z Axis	-3.1707e-004 in/in
XY Shear	5.0506e-006 in/in
YZ Shear	3.6697e-007 in/in
XZ Shear	-2.8492e-007 in/in
Equivalent (von-Mises)	1.057e-003 in/in
Maximum Principal	1.057e-003 in/in
Middle Principal	-3.1694e-004 in/in
Minimum Principal	-3.1734e-004 in/in
Intensity	1.3743e-003 in/in

Maximum Value Over Time	
Normal - X Axis	1.0563e-003 in/in
Normal - Y Axis	-3.1688e-004 in/in
Normal - Z Axis	-3.1689e-004 in/in
XY Shear	6.2583e-007 in/in
YZ Shear	6.1232e-008 in/in
XZ Shear	1.7109e-007 in/in
Equivalent (von-Mises)	1.0563e-003 in/in
Maximum Principal	1.0563e-003 in/in
Middle Principal	-3.1685e-004 in/in
Minimum Principal	-3.1692e-004 in/in
Intensity	1.3732e-003 in/in

The FEM analysis produced three important results for testing and evaluating the performance of the final torque wrench design. The maximum normal stress for the model was 35.7 ksi and happened near the drive transition. The load-point deflection under the 600 in-lbf torque was 0.0694 in which shows that the handle was stiff but still allowed bending strain. The strain at the gauge location was 1,057 microstrain which is better than expected when it comes to achieving the required electrical output.

7. Torque wrench sensitivity in mV/V using strains from FEM analysis

Further probing of strain at the point of the center of the strain gauge on the handle allows us to see the exact strain at the location of the gauge. The details for maximum values of strain at the probed location are shown in Figure 5 for both before and after mesh refinement. Note that the strain gauge is oriented along the x-axis given the wrench geometry in Ansys.

Given the values of the Normal -X Axis max strain over time, we can calculate the sensitivity of our wrench using the output of a half-bridge strain gauge:

Pre mesh refinement:

$$\text{Gauge output} = 2k\varepsilon/4 = \varepsilon \times 10^3 \text{ mV/V} = 1.057 \text{ mV/V}$$

Post mesh refinement:

$$\text{Gauge output} = 2k\varepsilon/4 = \varepsilon \times 10^3 \text{ mV/V} = 1.0563 \text{ mV/V}$$

Both versions of the mesh produce an output/sensitivity that meets the design requirements ($> 1 \text{ mV/V}$)

8. Strain gauge selected (give type and dimensions). Note that design must physically have enough space to bond the gauges